Trade Store

High-Level Design and Architecture

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Version	1.0, 12 November 2026
Status	Implemented through phase 8; phase 9 (load run and tuning) on hold
Source	github.com/vijaydixitffis/dws-trade-app
Stack	Java 21, Spring Boot 4.1, Apache Kafka, PostgreSQL, MongoDB, React

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1. Problem statement

The assignment asks for a Java application using Spring Boot and a streaming tool, with both a SQL and a NoSQL database. It must be built test-first with JUnit, delivered through a pipeline (GitHub Actions preferred) that runs regression tests and an open-source vulnerability scan, and the build must fail on critical or blocker vulnerabilities. PlantUML diagrams are requested where feasible.

The business problem is stated as follows.

> *Thousands of trades are being sent to a single store, through any method of transmission. The objective is a trade store that organises and stores each trade in a specific sequence.*

Each trade carries: Trade Id, Version, Counter-Party Id, Book-Id, Maturity Date, Created Date and an Expired flag (Y or N). The store must enforce three rules.

#	Rule from the assignment
1	A trade with a lower version than the stored one is rejected with an exception. A trade with the same version replaces the current record.
2	A trade whose maturity date is earlier than today is rejected.
3	When a trade's maturity date has passed, the store automatically marks it as expired.

The sample table in the assignment shows several versions of the same trade stored side by side (Trade T2 appears at version 2 and version 1), which tells us that the store keeps history rather than only the latest version.

What makes it hard

- **Volume and concurrency.** Many trades arrive at once, from any source, and several versions of one trade may be in flight together.
- **Order sensitivity.** Rule 1 depends on the order trades arrive in; a lower version arriving late must not overwrite a higher one, even when two processors work in parallel.
- **Time-based behaviour.** Rules 2 and 3 depend on "today", and rule 3 must act without any trade arriving.

- **No lost trades.** A trade accepted for processing must end up either stored or rejected with a recorded reason.

2. Scope, goals and decisions

In scope

- A trade capture API, a streaming backbone, a trade processor that enforces the three rules, a SQL trade store, and a NoSQL history of outcomes.
- Additions requested by the sponsor: a trade posting simulator and a dashboard UI.
- A test-first build, a delivery pipeline with a vulnerability gate, and PlantUML diagrams.

Out of scope for this release

Authentication and authorisation, TLS, multi-broker Kafka and database replication in the supplied Compose file, Kubernetes manifests, and measured load-test figures (phase 9).

Decisions agreed with the sponsor

Topic	Decision
Streaming tool	Apache Kafka, one container with no extra coordination service, so it can be demonstrated locally, in the pipeline and at a client site with Docker alone.
Databases	PostgreSQL as the system of record; MongoDB for the history of store changes and rejections.
Packaging	Separate Spring Boot services in one Maven multi-module repository, plus a React dashboard.
Rejections	Domain exception, a rejected-trades topic, stored in MongoDB with the reason, shown on the dashboard; the capture API also answers HTTP 422 at once when it can decide alone (past maturity).
Version history	Every version of a trade is kept, one row per (trade id, version).
"Today"	The date in the server's own time zone; an administrator may configure another zone.
Maturing today	Accepted. Only a date earlier than today is rejected (default, not objected to).
Created Date on replace	Set to the date the replacing trade was received (default, not objected to).
Sample dates in the assignment	The 2020 sample rows are ignored; all tests use dates relative to today.
UI style	Plain corporate palette of blue, grey and white.

3. Solution overview and components

Trades enter through a stateless capture API and are placed on a Kafka topic keyed by trade id. A processor consumes the topic, applies the rules against PostgreSQL in a single transaction, and announces every outcome on Kafka. A separate query service turns those announcements into a MongoDB history and feeds the dashboard. The simulator and the dashboard sit at the top.

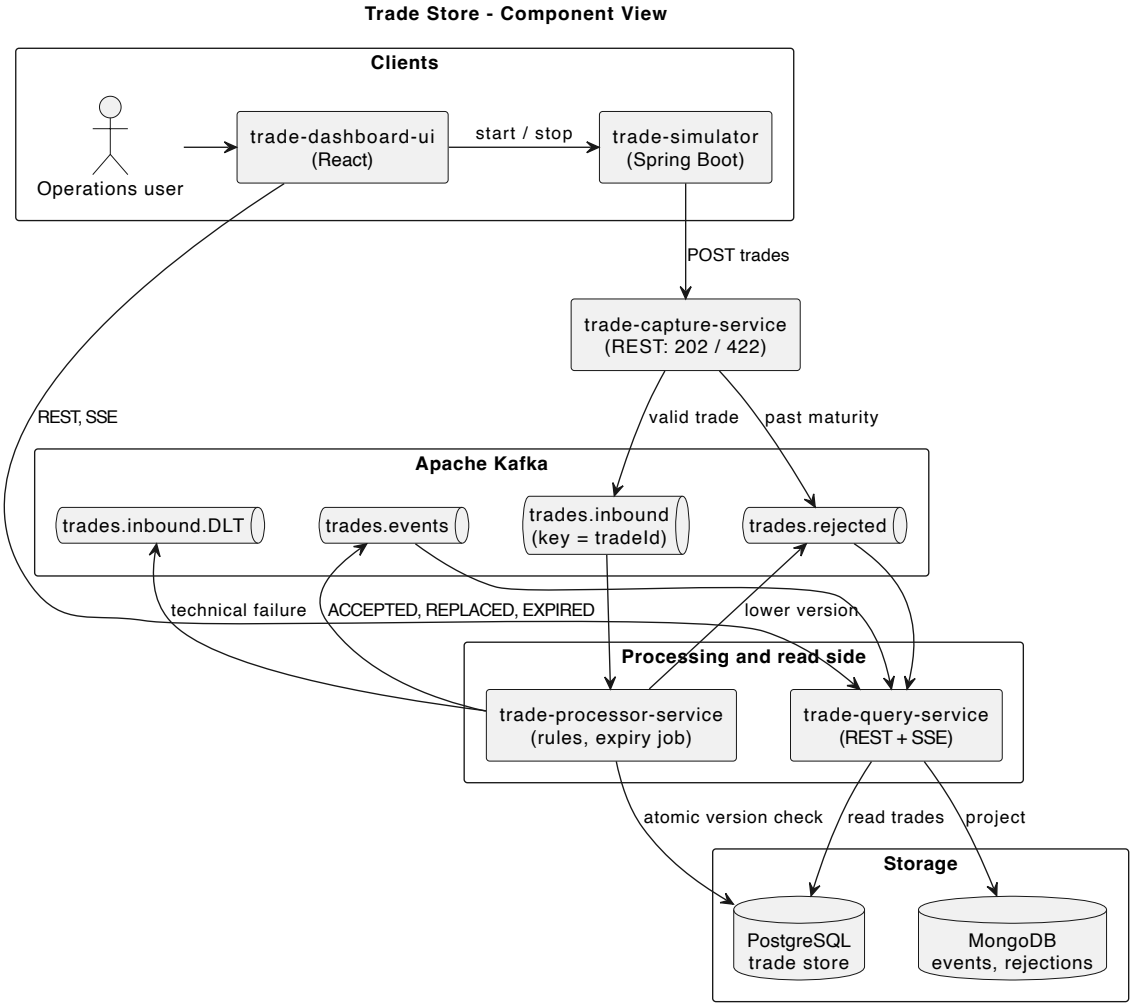


Figure 1. Component view

Component	Responsibility	State
trade-simulator	Generates trades at a chosen rate and mix (new, higher, same and lower version, past maturity) and posts them to the capture API. A run can also fix the counter-party and book pools, the maturity window and the trade id prefix. Started and stopped through a REST API or the dashboard, which offers a quick run dialog and a detailed run dialog.	None
trade-capture-service	Receives single and batch trades, validates structure, applies the maturity rule (HTTP 422 on failure, with the rejection recorded), and publishes accepted trades to `trades.inbound`. Answers 202 only after Kafka acknowledges the record.	None
Kafka	Topics `trades.inbound`, `trades.events`, `trades.rejected` and `trades.inbound.DLT`.	Log

trade-processor-service	Consumes the inbound topic, enforces the version and maturity rules in one database transaction, publishes the outcome, retries technical failures and parks unreadable records. Runs the expiry job.	PostgreSQL
PostgreSQL	The trade store.	-
trade-query-service	Projects outcome topics into MongoDB, serves the trade list from PostgreSQL and the history from MongoDB, and pushes live updates to the dashboard over Server-Sent Events.	MongoDB
MongoDB	History of every store change and every rejection.	-
trade-dashboard-ui	Summary counters, the trade store in sequence, rejected trades with reasons, and simulator controls: a quick run dialog (bulk mix) and a detailed run dialog (mix sliders and trade variety).	None

> **Why a separate query service.** The processor's only job is to keep up with the inbound stream. Dashboard reads and browser connections run elsewhere, so they cannot slow it down.

> **Why the capture API does not write to the database.** It stays stateless and fast, Kafka absorbs bursts, and the processor can be stopped or redeployed without losing accepted trades.

4. Technology choices

Concern	Choice	Why
Language and framework	Java 21, Spring Boot 4.1.2 (Spring Framework 7, Tomcat 11)	Required by the assignment. Boot 4 was adopted because the 3.5 line carried critical Tomcat and Spring MVC vulnerabilities that failed the pipeline's own gate.
Streaming	Apache Kafka 3.9, single node, KRaft mode	Partitioning by key gives per-trade ordering; replayable log; runs from one container.
SQL database	PostgreSQL 17	ACID transactions and an atomic conditional upsert (`INSERT ... ON CONFLICT ... WHERE`) that enforces the version rule safely under concurrency; row locks that skip locked rows for the expiry job.
NoSQL database	MongoDB 8.2	Append-style history and audit documents of varying shape (events and rejections), keyed by event id so redelivery overwrites itself.
Schema management	Flyway	Versioned, repeatable database schema.
Scheduled-job lock	ShedLock on PostgreSQL	Only one processor instance runs the expiry job.
UI	React, Vite, TypeScript, plain CSS	Quick to build and run; no UI framework needed.
Tests	JUnit 5, AssertJ, Mockito, Testcontainers, Awaitility, JaCoCo; Vitest and Testing Library for the UI	Real Kafka, PostgreSQL and MongoDB in tests rather than mocks of them.
Delivery	GitHub Actions; Trivy, `npm audit` and OWASP Dependency-Check	Named in the assignment; fails on critical findings.
Packaging	Docker and Docker Compose	One command starts the whole system anywhere Docker runs.

No separate application server is installed: each Spring Boot service embeds Tomcat.

5. Runtime flows

The main flow covers all three outcomes for a trade: accepted or replaced, rejected for a past maturity date, and rejected for a lower version.

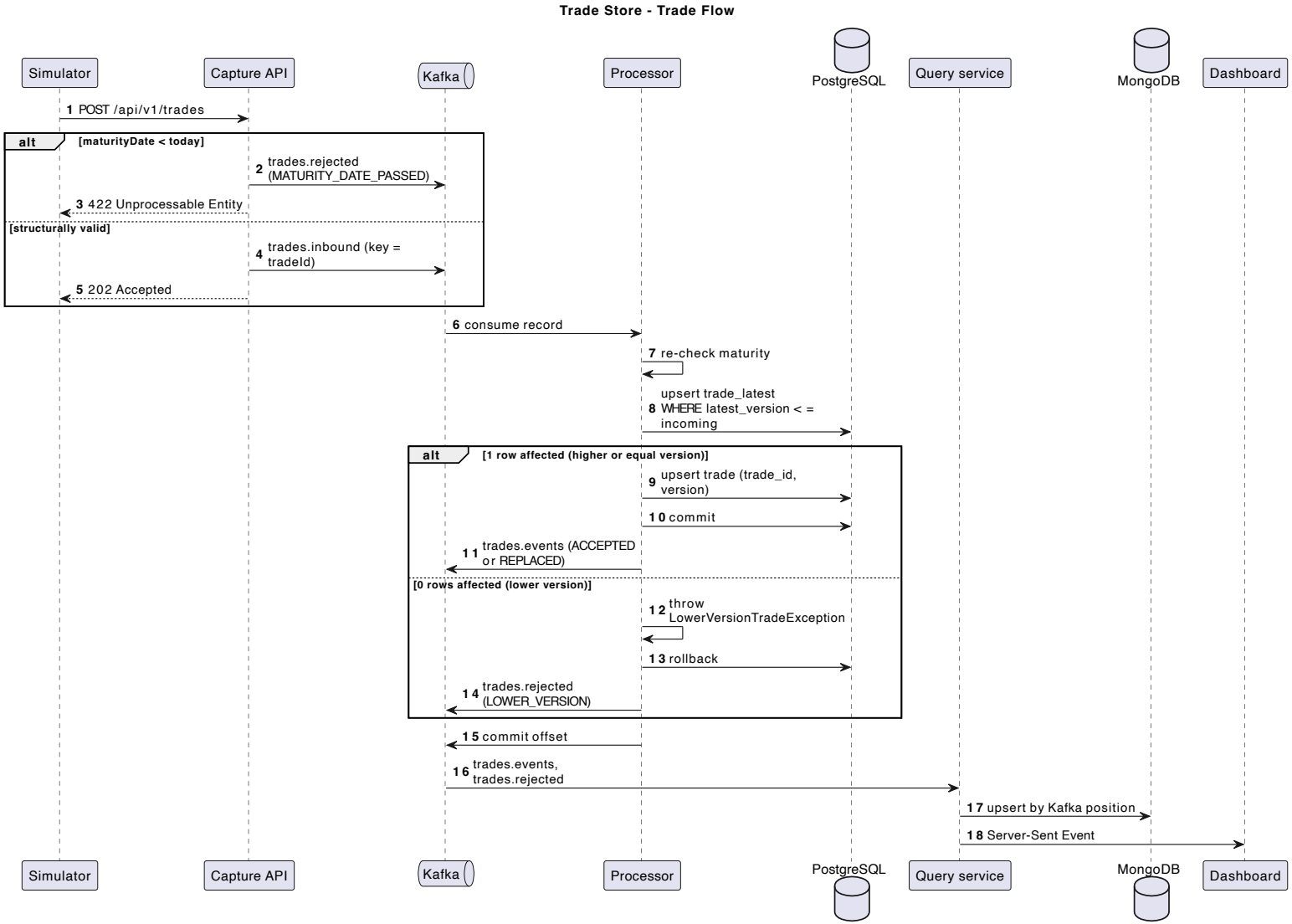


Figure 2. Trade flow from capture to dashboard

5.1 Reading the trade flow

1. The simulator (or any client) posts a trade, or a batch of up to 1,000, to the capture API.
2. A maturity date earlier than today is answered with HTTP 422, and the rejection is published to `trades.rejected` so it is stored and visible. A malformed trade is answered with 400. Otherwise the trade is published to `trades.inbound` keyed by trade id and the caller receives 202 once Kafka has acknowledged it.
3. The processor consumes the record. It checks maturity again, because other producers may write to the topic directly ("any method of transmission"), then applies the version rule inside one database transaction.
4. If the store accepts the trade it publishes an `ACCEPTED` event (new row) or `REPLACED` event (same version). If the rule is broken, the store raises `LowerVersionTradeException` or `MaturityDatePassedException`, the transaction rolls back, and a rejection with the reason is published to `trades.rejected`.
5. The Kafka offset is committed only after the database commit and the outcome publish have completed.
6. The query service consumes both outcome topics, stores them in MongoDB keyed by event id, and pushes them to connected dashboards.

A business rejection is a normal outcome and goes to `trades.rejected`. A technical failure (a message that cannot be read, or the database unavailable after retries) goes to `trades.inbound.DLT`. Keeping them apart lets operations tell bad trades from broken infrastructure.

5.2 Automatic expiry

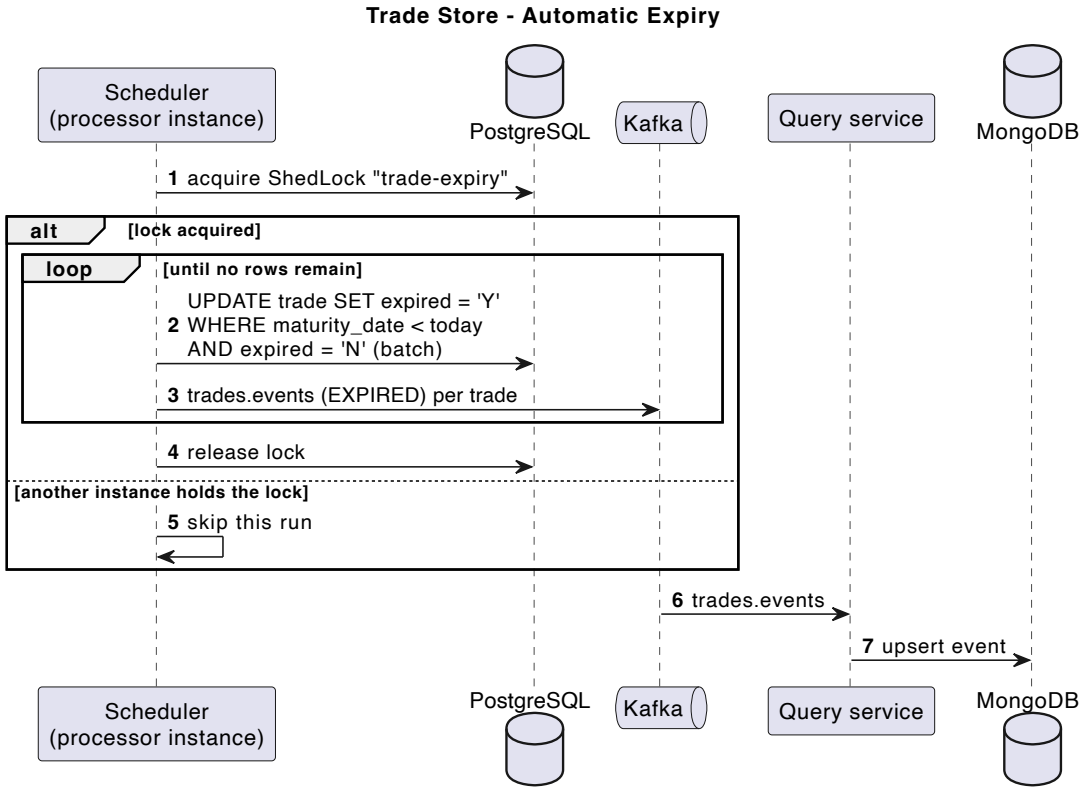


Figure 3. Expiry job

The job runs shortly after midnight in the business zone and once at startup, so trades that matured while the service was down are caught. A database lock allows one run at a time across processor instances. It marks trades in batches of 500 and publishes one `EXPIRED` event per trade.

6. Design of the three rules

Rule 1: version

Incoming version compared with the highest stored	Result
Higher, or first for that trade id	A new row is added; earlier versions are kept.
Equal	The existing row is replaced.
Lower	Rejected with `LowerVersionTradeException`; nothing is changed.

The check is made by the database, not in application memory, so it is correct with any number of processor instances. A small table holds the highest version per trade. One statement claims the version and affects no row when a higher version is already recorded:

```
INSERT INTO trade_latest (trade_id, latest_version) VALUES (?, ?)
ON CONFLICT (trade_id) DO UPDATE SET latest_version = EXCLUDED.latest_version
WHERE trade_latest.latest_version <= EXCLUDED.latest_version;
-- 0 rows affected: a higher version exists, so reject
-- 1 row affected: upsert the trade row in the same transaction
```

The statement takes a row lock on that trade, so concurrent writers of the same trade queue up and each sees the other's result. Kafka's per-key ordering means this rarely matters in normal running; the guard keeps the rule correct if ordering is ever disturbed by a rebalance, a replay or a second producer.

Rule 2: maturity date

A trade is rejected when its maturity date is earlier than today; a trade maturing today is accepted. The rule lives in one shared class used by the capture API, the processor and the expiry job, so all three agree on the boundary. "Today" is read from an injected clock, which defaults to the server's own date and time zone and can be set by an administrator with `trade.business-zone` (environment variable `TRADE_BUSINESS_ZONE`). Tests fix the clock, including a case where the date differs between two zones.

Rule 3: expiry

The expiry job runs this against rows that are not yet expired, in batches, while skipping any rows currently locked by trade processing:

```
UPDATE trade SET expired = 'Y'
WHERE (trade_id, version) IN (
   SELECT trade_id, version FROM trade
   WHERE expired = 'N' AND maturity_date < :today
   ORDER BY trade_id, version LIMIT :batch FOR UPDATE SKIP LOCKED)
```

A partial index on maturity date for unexpired rows keeps the search fast as the store grows.

Store sequence

"Stored in a specific sequence" is implemented as the order of the sample table: trade id ascending, then version descending. The primary key and the query API both use it.

7. Data design

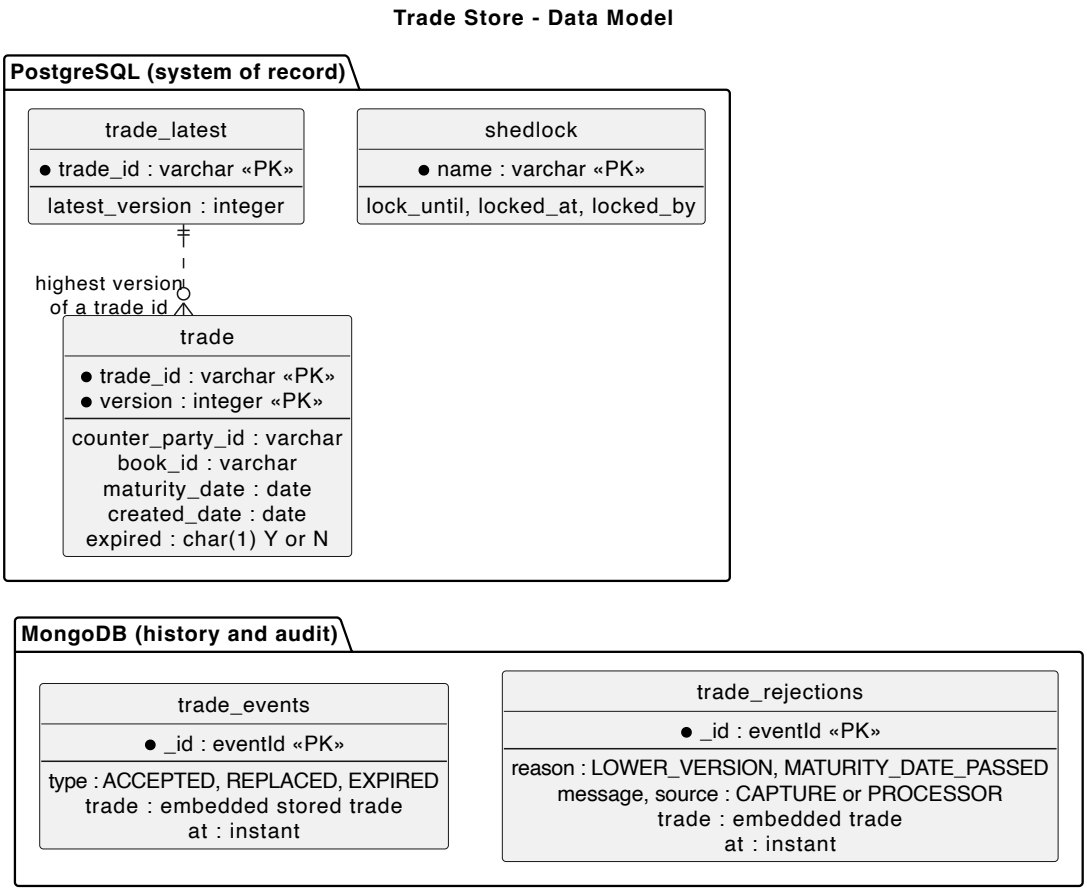


Figure 4. Data model

Store	Notes
PostgreSQL `trade`	Primary key (trade id, version). The expired flag is `Y` or `N`, as in the assignment. Dates are SQL dates.
PostgreSQL `trade_latest`	Highest version seen per trade id; the target of the atomic version guard.
PostgreSQL `shedlock`	Lock rows for the expiry job.
MongoDB `trade_events`	One document per store change (accepted, replaced, expired). The document id is the event id.
MongoDB `trade_rejections`	One document per rejection with reason, message, source (capture or processor) and the rejected trade.

Event ids for processed trades are built from the inbound Kafka position (topic, partition, offset). Redelivering the same record therefore overwrites the same MongoDB document instead of creating a duplicate. The schema is created by a Flyway migration.

Why two databases

PostgreSQL holds what must be correct: the current state of the store, changed under transactions. MongoDB holds what is read, counted and audited: a history that is easy to append to and to show on a dashboard without touching the store's tables.

8. Deployment view

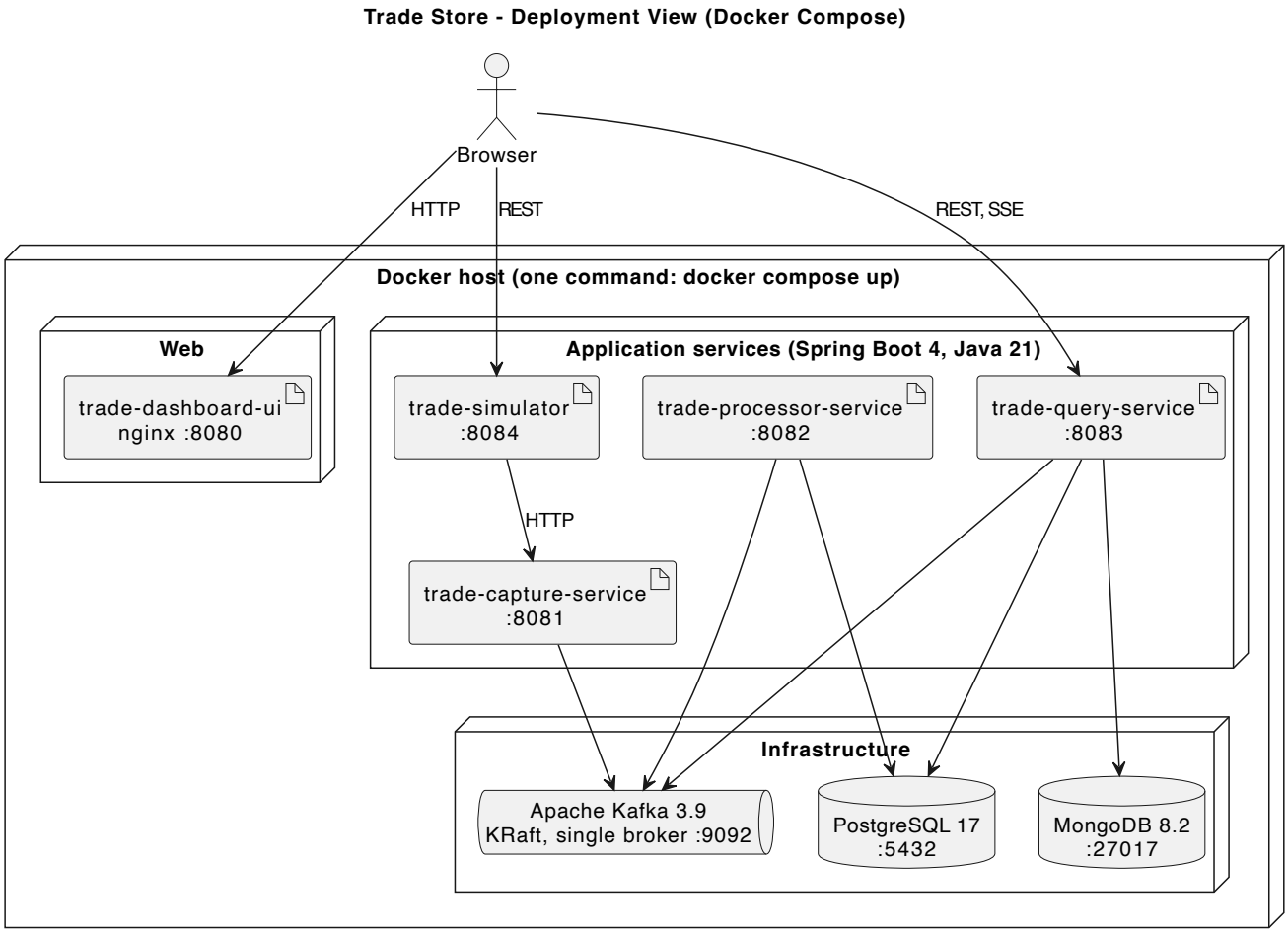


Figure 5. Deployment on Docker Compose

One command, `docker compose -f deploy/docker-compose.yml up --build`, builds and starts all eight containers with health checks and start-up ordering. Only Docker is required. The services read their settings from environment variables, so the same images move between a laptop, the pipeline and a client site.

Setting	Purpose
`TRADE_BUSINESS_ZONE`	Time zone that defines "today"; empty means the server's own zone.
`SPRING_KAFKA_BOOTSTRAP_SERVERS`	Kafka address.
`SPRING_DATASOURCE_URL`	PostgreSQL address.
`SPRING_MONGODB_URI`	MongoDB address.
`trade.inbound-partitions`, `trade.processor.concurrency`	Topic partitions (6) and consumer threads per processor instance (3).
`trade.expiry.cron`, `trade.expiry.batch-size`	Expiry schedule (00:01 in the business zone) and batch size (500).
`trade.retry.*`	Retry interval (1 s) and attempts (3) before a record is parked.

For production the same services run on any container platform. The Compose file's single Kafka broker and single database instances would be replaced by a three-broker cluster and a highly available PostgreSQL and MongoDB (section 9).

9. Non-functional requirements and how the solution meets them

The assignment names volume ("thousands of trades"), ordering, no loss and automatic behaviour. The table below turns those into explicit requirements and states how each is met. The last column is candid: **Met** means built and tested; **Partly** or **Not yet** marks work that remains.

Requirement	How the solution addresses it	Status
Throughput	Capture is stateless, runs on Java 21 virtual threads, offers a batch endpoint (up to 1,000 trades per call) and publishes through a batching, compressed producer (`linger.ms=5` , lz4). The inbound topic has 6 partitions with a consumer thread per partition (3 threads per instance by default). Each trade costs two short database statements in one transaction. The expiry job is batched and skips locked rows so it does not stall processing.	**Design supports it; figures not yet measured** (phase 9)
Scalability	Capture and query are stateless and scale by adding instances behind a load balancer. Processing scales by adding instances up to the partition count; more partitions raise that ceiling. Kafka absorbs bursts so producers and the processor run at different speeds. Consumer lag is the signal to scale.	**Met by design**
Ordering	Records are keyed by trade id, so all versions of a trade share a partition and are handled in order by one thread.	**Met**
Data consistency under concurrency	The version rule is a single atomic conditional statement with a row lock, so correctness does not depend on arrival order or on how many processors run. A test sends 40 shuffled versions of one trade from 8 threads and checks that the highest version always wins and the recorded highest equals the stored highest.	**Met**
No loss of accepted trades	The producer uses `acks=all` and idempotence. The capture API answers 202 only after Kafka acknowledges the record, and answers 503 if it cannot. The processor commits its offset only after the database commit and the outcome publish.	**Met** (single broker in Compose, see Availability)
Delivery guarantee and idempotency	Delivery is at-least-once. Reprocessing is harmless: the same version replaces itself, and MongoDB documents are keyed by Kafka position. After a crash between commit and publish, the trade is reprocessed and the event is published then, with type `REPLACED` rather than `ACCEPTED` .	**Met**, with that documented wrinkle
Fault handling	Technical failures are retried (3 times, 1 s apart), then the record is parked on `trades.inbound.DLT` so the partition keeps moving. Unreadable or incomplete records are parked at once. Business rejections never use the dead-letter topic.	**Met**
Availability	Capture, processor and query are stateless and restart without losing data; Kafka buffers trades while the processor is down, and capture keeps accepting. Containers restart on failure and have health checks. The supplied Compose file runs a single broker and single databases, so it is not highly available.	**Partly**: production needs 3 brokers (replication factor 3, min in-sync replicas 2) and replicated databases
Durability	Trades are committed to PostgreSQL before the offset moves; inbound records stay in Kafka until retention removes them, allowing replay.	**Met** (replication is a deployment choice)
Auditability	Every store change and every rejection, with its reason and source, is kept in MongoDB and shown on the dashboard; Kafka topics allow replay.	**Met**

Correctness of time	One injected clock supplies "today" to capture, processor, expiry and simulator; the zone is the server's, or the administrator's override. The shared maturity rule defines the boundary once. A test shows a Tokyo clock moves the date a day ahead of UTC.	**Met**
Configurability	Time zone, partitions, concurrency, retry, expiry schedule, batch size and allowed browser origins are settings, not code.	**Met**
Observability	Spring Actuator health, info and metrics on every service. The processor counts outcomes (`trades.processed` by outcome: accepted, replaced, lower version, maturity passed). Rejections are logged with the reason. The dashboard shows live counters and recent rejections.	**Partly**: no distributed tracing or correlation id yet, and no metrics exporter configured
Security	Inputs are validated (required fields, version of at least 1, batch size limit). Containers run as a non-root user. Browser access is limited to an allow-list of origins. Dependencies are scanned on every build and critical findings fail it.	**Not yet**: no authentication, authorisation or TLS, and the Compose file uses default database credentials. These must be added before any non-demo use.
Maintainability and testability	Small modules with one job each; the rules live in a shared module with no framework dependency; every rule was written test-first. 63 backend tests, 11 UI tests and a 17-check end-to-end script. Line coverage above 90 % in every module, with a build-failing gate at 80 %.	**Met**
Portability and ease of setup	Docker is the only requirement to run the system; the same Kafka and database images are used in tests, the pipeline and the demo. The code imports into Eclipse as ordinary Maven projects.	**Met**
Supply-chain safety	Trivy and `npm audit` run on every build; any critical finding (CVSS 9.0 or above) fails it. OWASP Dependency-Check runs too when an NVD API key is configured. The gate has already proved useful: it forced the move to Spring Boot 4.	**Met** (OWASP run pending the key)
Data growth	Kafka retention bounds the log. The partial index keeps expiry cheap. MongoDB history grows without a limit today.	**Partly**: add a retention (time-to-live) policy for history
Usability	The dashboard shows the store in the assignment's column order and date format, live counters, rejection reasons, and simulator controls, in a plain corporate style.	**Met**

10. Quality, testing and delivery pipeline

Test-first approach

For each backend phase the tests were written and committed first (they failed), then the implementation was written until they passed; the commit history shows `test:` commits before the matching `feat:` commits. The dashboard is the exception: its tests and components were written together.

Level	What it covers	Count
Rule unit tests	Version and maturity boundaries, exceptions, business clock	8
Store tests on real PostgreSQL	Lower, same and higher version, maturity boundary, atomic rollback, 8-thread concurrency	8

Processor and expiry	One outcome per trade, zone-dependent date, expiry boundary, batches, re-run safety	9
Kafka integration	Accept, replace, reject in arrival order; past maturity; unreadable and incomplete records parked while good ones continue; capture API publishing with the right key	4
HTTP contracts	202, 422, 400, 503, batch; simulator run control and validation of its variation inputs	15
Query side	Store ordering and paging, MongoDB projection and de-duplication, summary, live feed, CORS	11
Simulator generator	Scenarios are true to the versions already sent; mix matches the request; chosen counter-parties, books, maturity window and id prefix are honoured	8
Dashboard	Table, date format, summary, rejections, filter, quick and detailed run dialogs (open, cancel, validation, request contents)	11
End-to-end regression	Whole system on Docker Compose, using dates relative to today	17 checks

Delivery pipeline

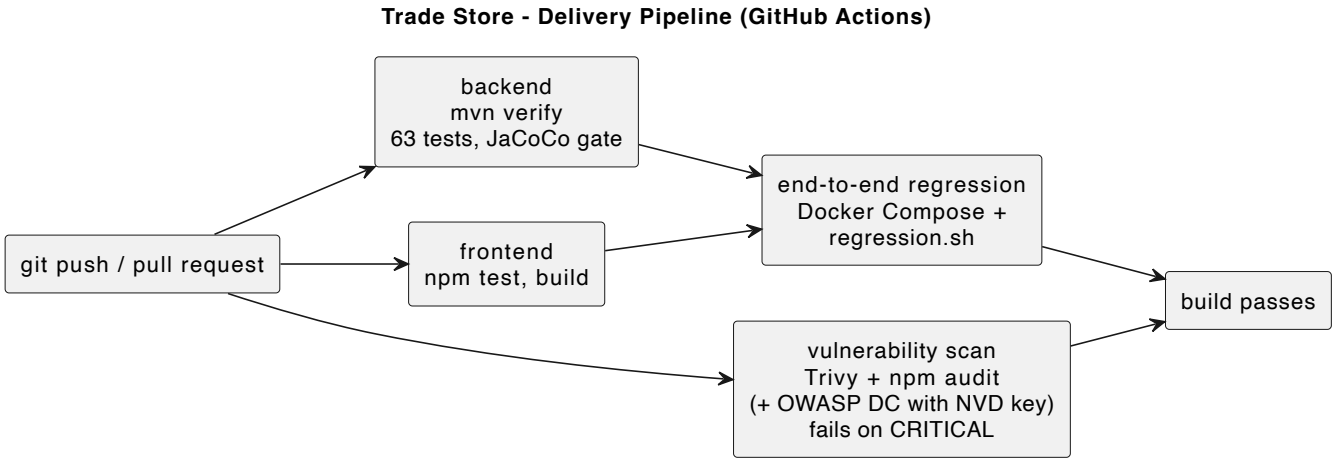


Figure 6. GitHub Actions pipeline

On every push and pull request the backend and dashboard jobs build and test, the vulnerability job scans in parallel, and the end-to-end regression runs once the build jobs pass. The latest run on the main branch has all four jobs green.

The same checks run without GitHub: `mvn verify`, `docker compose up --build -d` followed by `deploy/regression.sh`, and `mvn -Psecurity-scan verify` for OWASP Dependency-Check.

11. Risks, limitations and next steps

Item	Detail and proposed action
Throughput not yet measured	Phase 9 will run the simulator at increasing rates and record trades per second, consumer lag and latency, then tune partitions, threads and batching. On hold at the sponsor's request.
No authentication or TLS	Add OAuth2 or API-key access on the capture and query APIs, TLS on all links, and secrets from a vault instead of defaults.

Single broker and databases in Compose	Production deployment needs a Kafka cluster (replication factor 3, min in-sync replicas 2) and replicated PostgreSQL and MongoDB.
Commit-then-publish window in the processor	A crash between the database commit and the outcome publish makes the trade reprocessed, so its event is published as `REPLACED`. A transactional outbox would remove the wrinkle if exact event types matter.
Fixed partition count	Six partitions cap processor parallelism at six threads per consumer group; increasing the count later is possible but needs care for keyed ordering.
Live feed with several query instances	An event reaches only the browsers connected to the instance that consumed it. The dashboard also polls every three seconds, so it stays correct.
Observability gaps	Add correlation ids from HTTP header through Kafka header to logs, and a Prometheus exporter with dashboards and alerts on consumer lag and dead-letter volume.
History retention	Add a time-to-live on MongoDB history.
OWASP Dependency-Check	Not yet run: it needs an NVD API key as a repository secret. Trivy covers the gate meanwhile.
Class diagrams	Planned with phase 9.

12. Traceability from requirements to design and tests

Requirement	Design element	Verified by
Spring Boot application	Five Spring Boot modules	Build and all tests
Streaming tool	Kafka topics between capture, processor and query	Kafka integration tests; regression script
SQL and NoSQL databases	PostgreSQL trade store; MongoDB history	Store, projection and query tests
Rule 1: lower version rejected, same replaces	Atomic guard in the trade store; exceptions	Store tests, concurrency test, Kafka integration, regression
Rule 2: past maturity rejected	Shared maturity rule in capture and processor	Rule tests, HTTP 422 test, Kafka integration, regression
Rule 3: matured trades expired	Scheduled expiry job with lock	Expiry tests
Thousands of trades, any transmission	Partitioned Kafka, batch API, stateless capture, validation in the processor	Simulator 500-trade regression; load figures pending
Stored in a specific sequence	Trade id ascending, version descending	Query repository test, dashboard test
JUnit, test-first	Tests committed before implementation	Commit history; 63 backend tests
Pipeline with regression and vulnerability scan; fail on critical	GitHub Actions workflow	Pipeline runs; the scan failed the build on real findings before the Boot 4 move
PlantUML diagrams	Component, sequence, data model, deployment and pipeline diagrams	Sources and rendered SVGs in `docs/diagrams`

Appendix: interfaces and configuration

Service (port)	Interface
Capture (8081)	`POST /api/v1/trades` returns 202, 422, 400 or 503. `POST /api/v1/trades/batch` accepts up to 1,000 trades and returns a result per trade.
Processor (8082)	Kafka consumer of `trades.inbound`; Actuator health, info, metrics.
Query (8083)	`GET /api/v1/trades` (paged, filter by trade id prefix), `/events`, `/rejections`, `/summary`, and `/stream` (Server-Sent Events).
Simulator (8084)	`POST /api/v1/simulations` starts a run (volume, scenario mix, and optional counterparties, books, maturity window and id prefix); `GET` and `DELETE /api/v1/simulations/current` report and stop it.
Dashboard (8080)	Web page served by nginx.

Repository layout: `trade-common` (model, rules, schema), `trade-capture-service`, `trade-processor-service`, `trade-query-service`, `trade-simulator`, `trade-dashboard-ui`, `deploy` (Compose, Dockerfile, regression script) and `docs`.